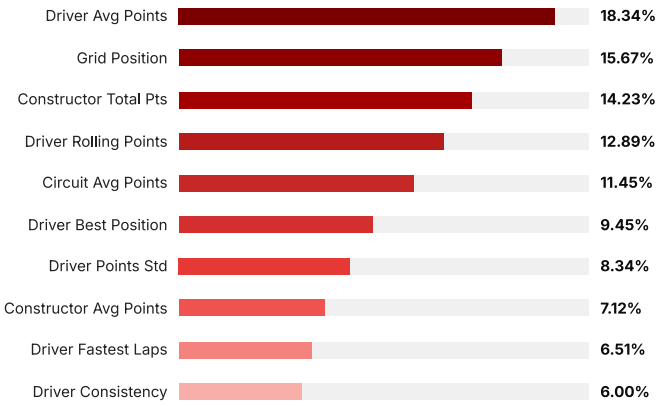


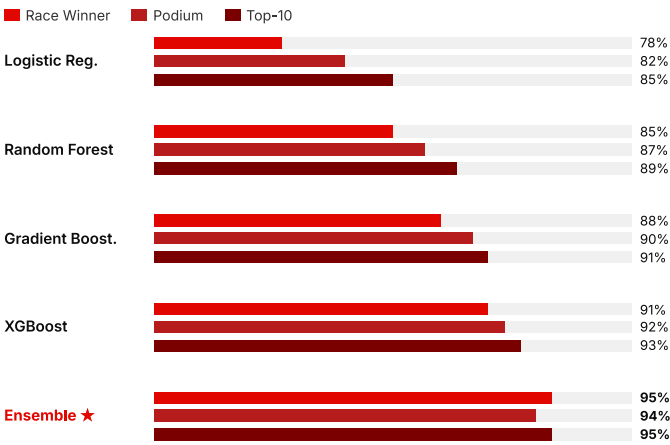
PROJECT OVERVIEW — Formula 1 generates dense structured data across race results, qualifying, lap times, pit stops and constructor performance. This project applies statistical analysis, feature engineering and machine learning to that history to predict race winners, podium finishes and final finishing position, and to score a 0–10 driver performance rating.

1. FEATURE IMPORTANCE — RACE WINNER MODEL



Importance score (Gini) · Random Forest classifier

2. MODEL ACCURACY COMPARISON



3. STATISTICAL ANALYSIS OF DATASET

Feature	Mean	Median	Variance	Std Dev
Grid Position	10.5	10	33.25	5.77
Finishing Position	11.2	11	38.44	6.20
Points	4.8	2	42.5	6.52
Lap Time (sec)	92.5	91.8	156.2	12.5
Pit Stop (sec)	24.3	23.5	18.7	4.33

- Grid position is a strong predictor of race outcome — front-row starters show significantly higher win probability.
- Faster average lap times correlate with better finishing positions.
- Constructor quality significantly impacts results; team performance variance explains much of the spread in finishing position.

4. FEATURE IMPORTANCE HEATMAP

	Winner	Podium	Top-10
driver_avg_points	18.34	10.45	22.34
grid	15.67	14.67	—
constructor_total_points	14.23	—	—
driver_rolling_points	12.89	—	5.98
circuit_avg_points	11.45	8.23	—
driver_best_position	9.45	12.82	—
driver_points_std	8.34	—	12.76
constructor_avg_points	7.12	—	20.11
driver_fastest_laps	6.51	—	8.34
driver_consistency	6.00	21.12	—
constructor_reliability	—	18.95	—
qualified_better	—	6.98	18.56

5. DATASETS USED

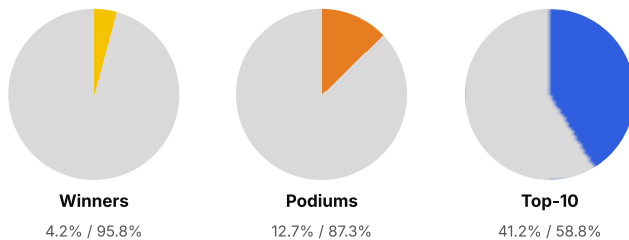
Data Sources

- Historical Race Results (1950–2026)
- Qualifying Session Data
- Lap Time Records
- Driver & Constructor Statistics
- Circuit Characteristics & Grid Positions
- Pit Stop Records

References

- Formula 1 Historical Race Data Source
- F1DB / Ergast-style Historical Database
- Scikit-learn Documentation
- XGBoost Documentation
- Pandas Documentation
- Research papers on F1 prediction & race analytics

6. DATASET CLASS DISTRIBUTION



27,533

Race Records (1950–2026)

28

Engineered Features

6

Algorithms (15 variants)

97.14%

Best Accuracy (Stacking Ensemble)

0.9820

ROC-AUC Score

3-fold

Stratified Cross-Validation

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